IE4728 Enterprise Network Design

Enterprise Network Design Project

Design Examples using Packet Tracer 8.0

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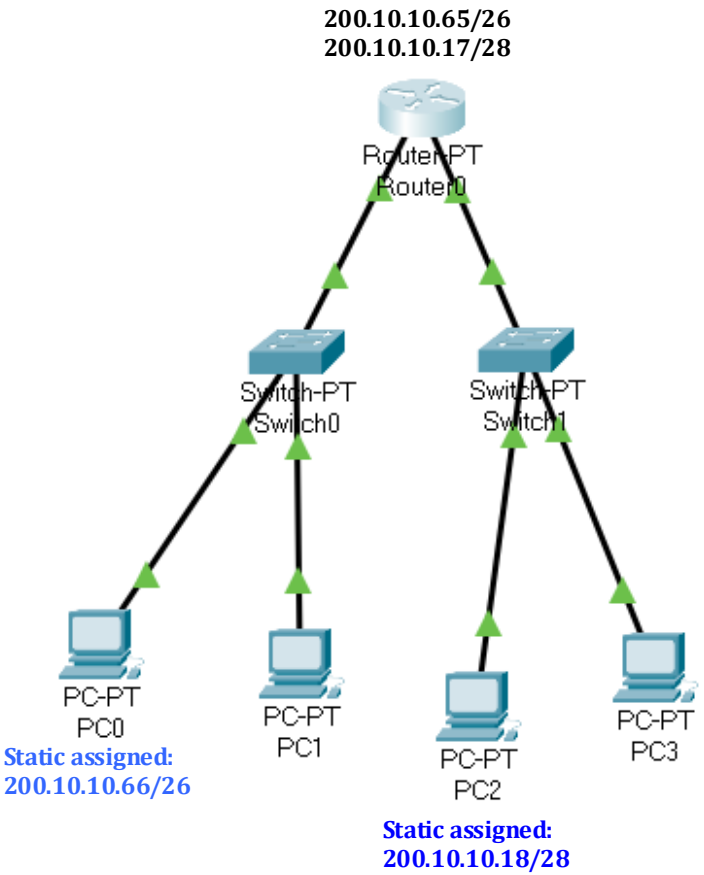
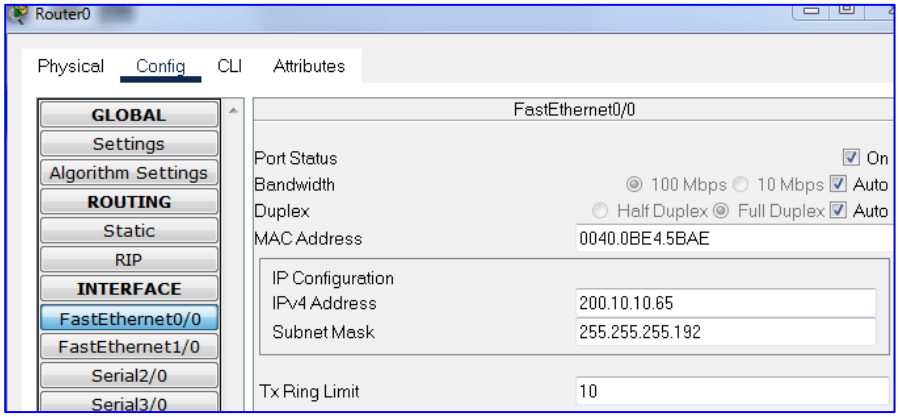
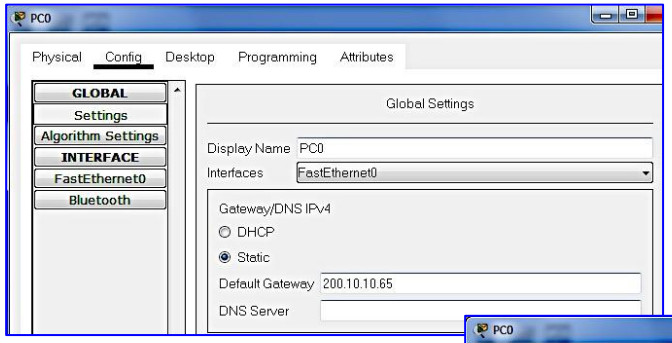
Static IP Configuration



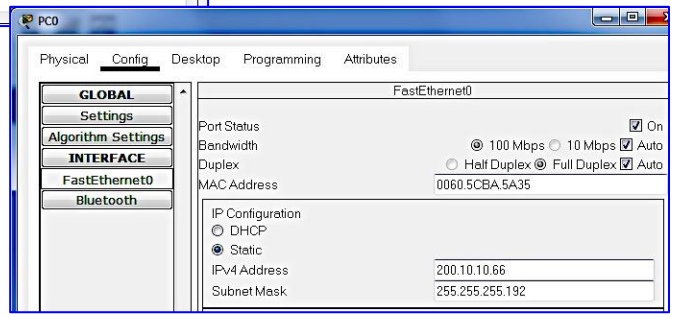


Router 0 interface address

← no shutdown



PC0 gateway address

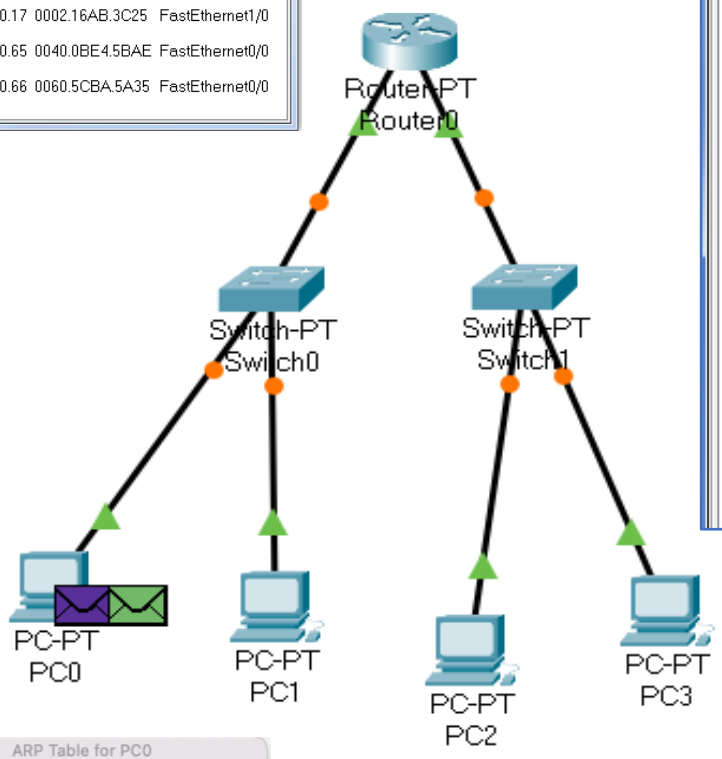


PC0 interface address

Static IP Simulation – ICMP/ARP events(1)

ARP Table for Router0

IP Address	Hardware Address	Interface
200.10.10.17	0002.16AB.3C25	FastEthernet1/0
200.10.10.65	0040.0BE4.5BAE	FastEthernet0/0
200.10.10.66	0060.5CBA.5A35	FastEthernet0/0



Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC0	ICMP
	0.000	–	PC0	ARP
	0.001	PC0	Switch0	ARP
	0.002	Switch0	Router0	ARP
	0.002	Switch0	PC1	ARP
	0.003	Router0	Switch0	ARP
	0.004	Switch0	PC0	ARP
	0.004	–	PC0	ICMP
	0.005	PC0	Switch0	ICMP
	0.006	Switch0	Router0	ICMP
	0.006	–	Router0	ARP
	0.007	Router0	Switch1	ARP
	0.008	Switch1	PC2	ARP
	0.008	Switch1	PC3	ARP
	0.009	PC3	Switch1	ARP
	0.010	Switch1	Router0	ARP

ARP resolving PC0 address when attempting PC0->PC3

Subsequent PC0 -> PC3

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	Router0	ICMP
	0.003	Router0	Switch1	ICMP
	0.004	Switch1	PC2	ICMP
	0.004	Switch1	PC3	ICMP
	0.005	PC3	Switch1	ICMP
	0.006	Switch1	Router0	ICMP
	0.007	Router0	Switch0	ICMP
	0.008	Switch0	PC0	ICMP

ARP Table for PC0

IP Address	Hardware Address	Interface
200.10.10.65	0040.0BE4.5...	FastEthern...

Status of Packet transmission:

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC0	PC3	ICMP	

Static IP Simulation – ICMP/ARP events(2)

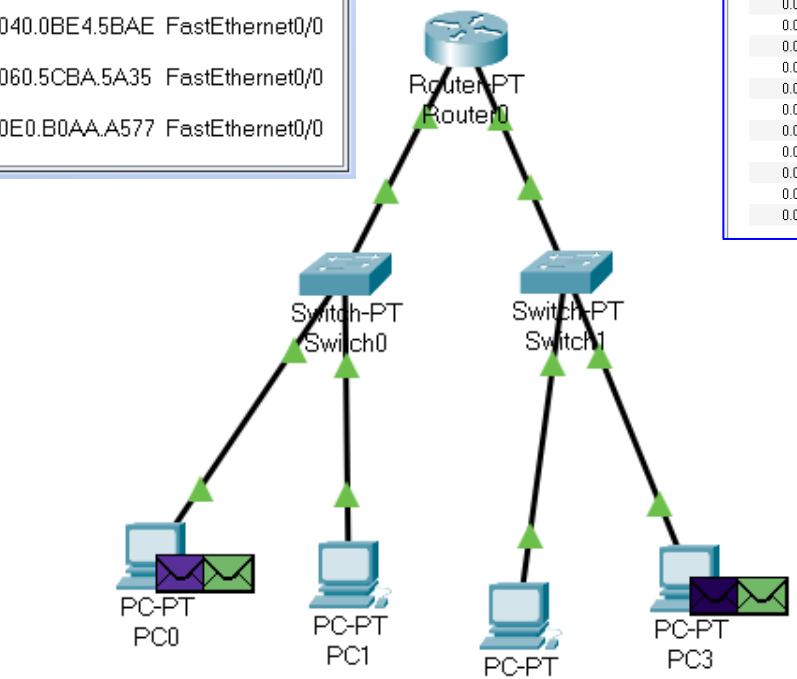
ARP Table for Router0

IP Address	Hardware Address	Interface
200.10.10.17	0002.16AB.3C25	FastEthernet1/0
200.10.10.19	000D.BDB7.9895	FastEthernet1/0
200.10.10.65	0040.0BE4.5BAE	FastEthernet0/0
200.10.10.66	0060.5CBA.5A35	FastEthernet0/0
200.10.10.67	00E0.B0AA.A577	FastEthernet0/0

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC0	ICMP
	0.000	–	PC0	ARP
	0.000	–	PC3	ICMP
	0.000	–	PC3	ARP
	0.001	PC0	Switch0	ARP
	0.001	PC3	Switch1	ARP
	0.002	Switch0	Router0	ARP
	0.002	Switch0	PC1	ARP
	0.002	Switch1	Router0	ARP
	0.002	Switch1	PC2	ARP
	0.003	Router0	Switch0	ARP
	0.003	Router0	Switch1	ARP
	0.004	Switch0	PC0	ARP
	0.004	Switch1	PC3	ARP
	0.004	–	PC0	ICMP
	0.004	–	PC3	ICMP
	0.005	PC0	Switch0	ICMP
	0.005	PC3	Switch1	ICMP

ARP resolving PC0 and PC3 addresses on transmission of PC0->PC3 and PC3->PC1



ARP Table for PC0

IP Address	Hardware Address	Interface
200.10.10.65	0040.0BE4.5...	FastEthern...

ARP Table for PC3

IP Address	Hardware Address	Interface
200.10.10.17	0002.16AB.3C25	FastEthernet0

Subsequent ICMP packets transmissions use mac addresses already in ARP table

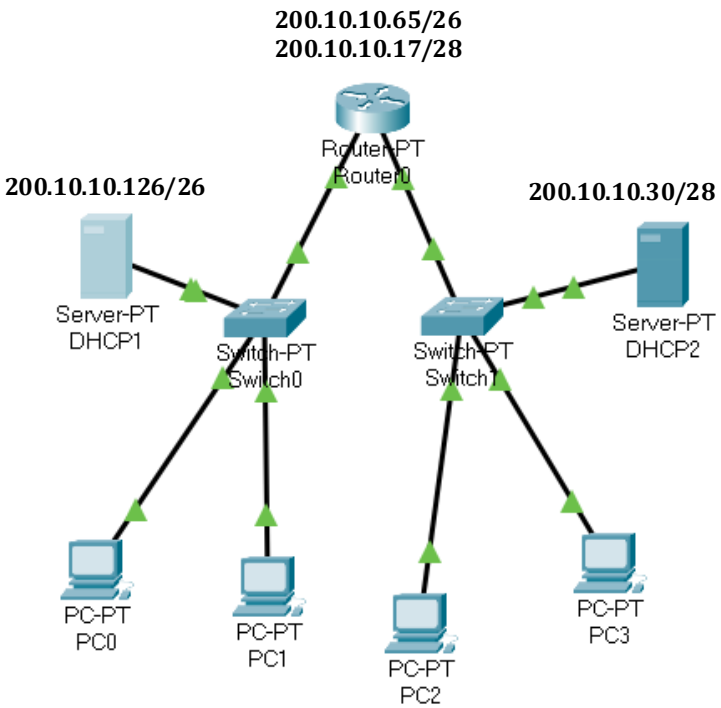
Event List

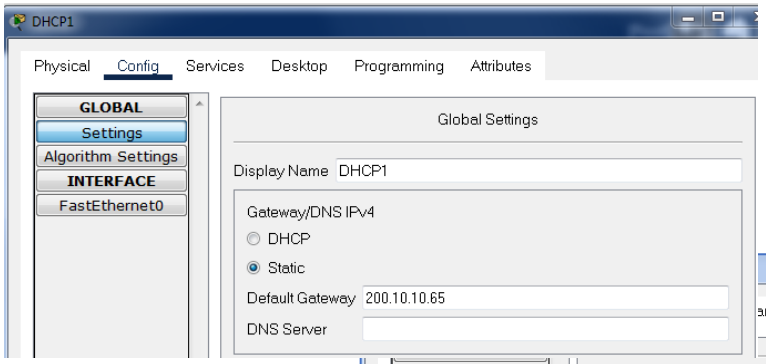
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC0	ICMP
	0.000	–	PC3	ICMP
	0.001	PC0	Switch0	ICMP
	0.001	PC3	Switch1	ICMP
	0.002	Switch0	Router0	ICMP
	0.002	Switch1	Router0	ICMP
	0.003	Router0	Switch1	ICMP
	0.003	Router0	Switch0	ICMP
	0.004	Switch1	PC3	ICMP
	0.004	Switch0	PC1	ICMP
	0.005	PC3	Switch1	ICMP
	0.005	PC1	Switch0	ICMP
	0.006	Switch1	Router0	ICMP
	0.006	Switch0	Router0	ICMP
	0.007	Router0	Switch0	ICMP
	0.007	Router0	Switch1	ICMP
	0.008	Switch0	PC0	ICMP
	0.008	Switch1	PC3	ICMP

Status of Packet transmission:

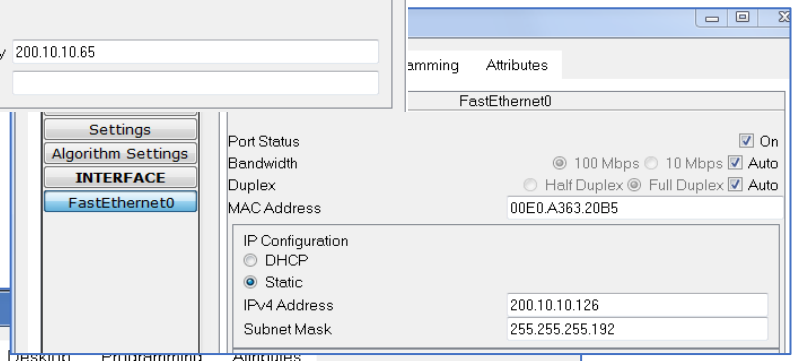
Fire	Last Status	Source	Destination	Type	Color
	Successful	PC0	PC3	ICMP	
	Successful	PC3	PC1	ICMP	

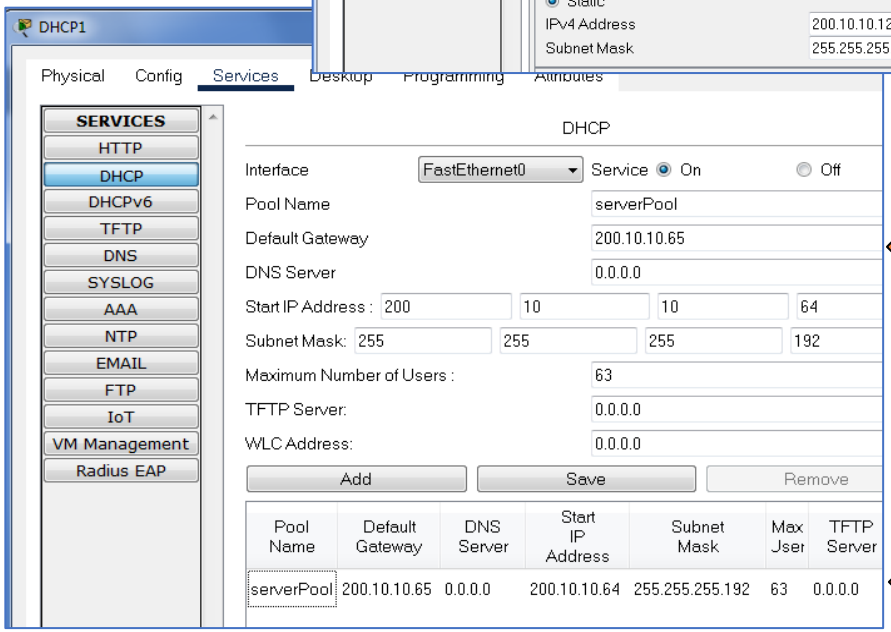
DHCP Host & Service Configuration





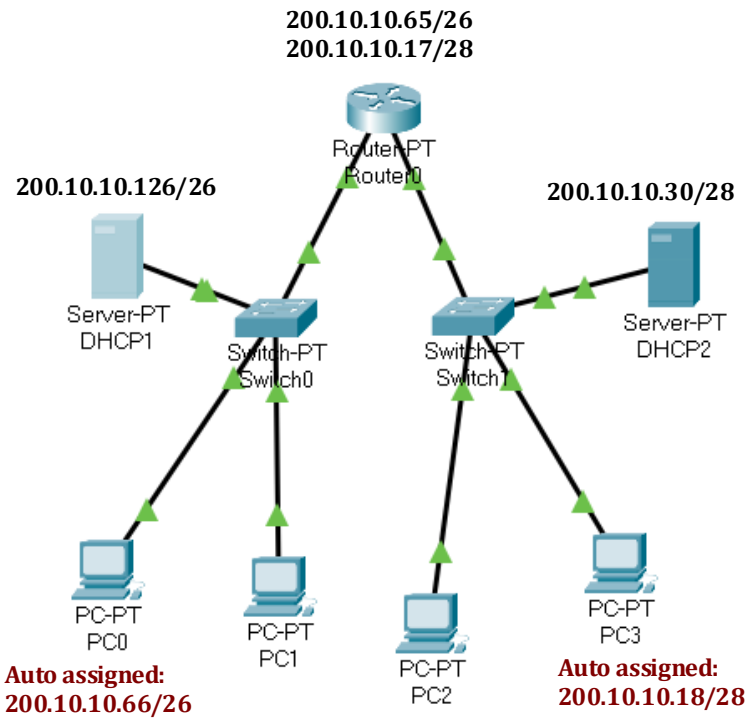
DHCP Gateway & Host IP config



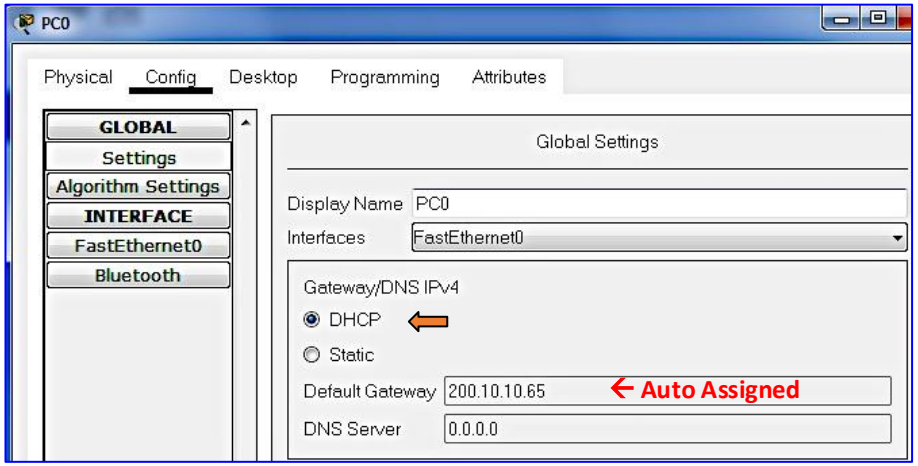


DHCP Service config

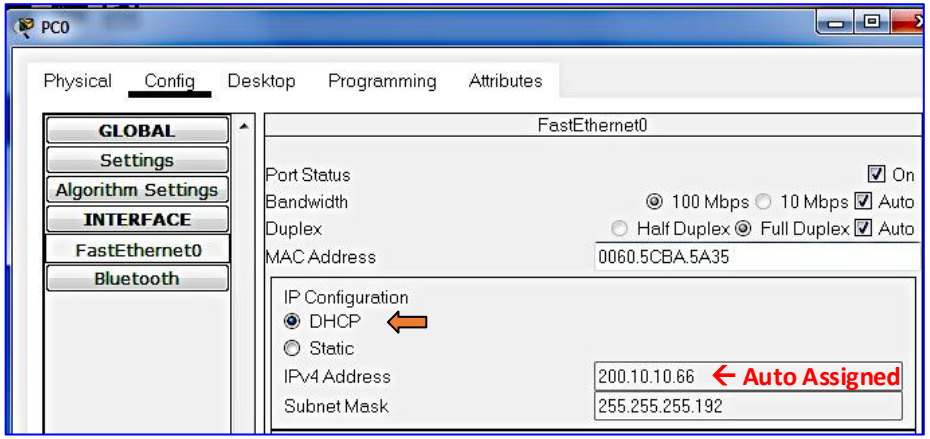
PC Host Configuration using DHCP



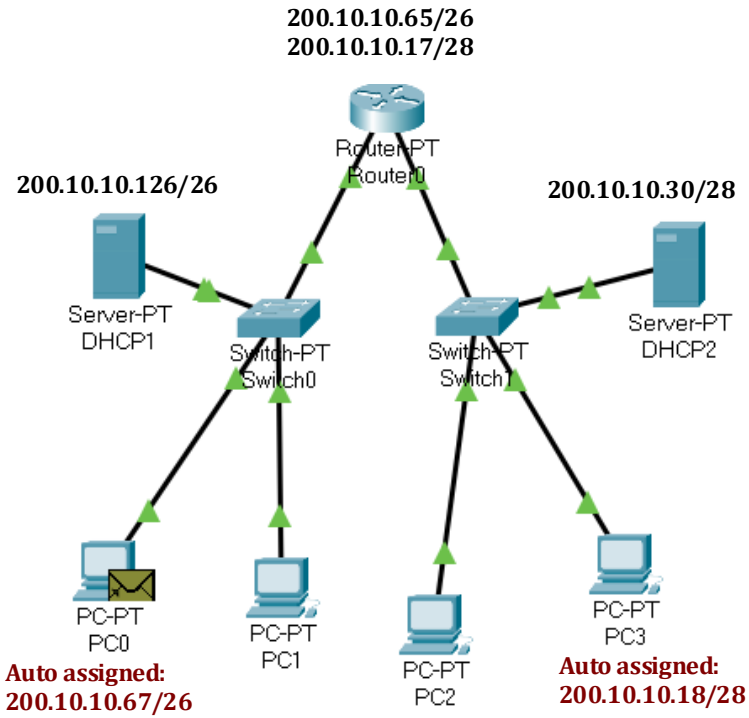
Select DHCP to set Default DHCP Gateway address on PC host



Dynamic host IP address automatically assigned using DHCP



Simulation - Dynamic Host IP



ICMP events

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	Router0	ICMP
	0.003	Router0	Switch1	ICMP
	0.004	Switch1	PC2	ICMP
	0.004	Switch1	PC3	ICMP
	0.004	Switch1	DHCP2	ICMP
	0.005	PC3	Switch1	ICMP
	0.006	Switch1	Router0	ICMP
	0.007	Router0	Switch0	ICMP
	0.008	Switch0	PC0	ICMP

Simulation status

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC0	PC3	ICMP	

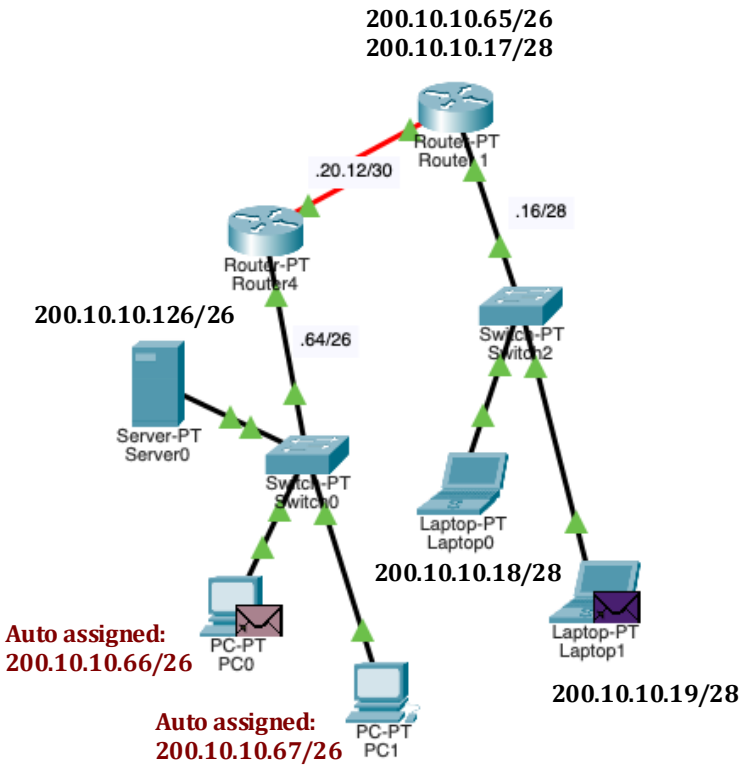
Out Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 200.10.10.67, Dest. IP: 200.10.10.18 ICMP Message Type: 8
Layer 2: Ethernet II Header 0060.5CBA. 5A35 >> 0040.0BE4.5BAE
Layer 1: Port(s): FastEthernet0

Impt: Re-assign packet and reset network card after changes to DHCP server

Note: PC0 was previously 200.10.10.66, and is being re-assigned with new IP 200.10.10.67 by DHCP

Simulation - Static & Dynamic Host IP



Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.000	--	Laptop1	ICMP
	0.001	PC0	Switch0	ICMP
	0.001	Laptop1	Switch2	ICMP
	0.002	Switch0	Router4	ICMP
	0.002	Switch2	Router 1	ICMP
	0.003	Router4	Router 1	ICMP
	0.003	Router 1	Router4	ICMP
	0.004	Router 1	Switch2	ICMP
	0.004	Router4	Switch0	ICMP
	0.005	Switch2	Laptop0	ICMP
	0.005	Switch2	Laptop1	ICMP
	0.005	Switch0	PC0	ICMP
	0.005	Switch0	PC1	ICMP
	0.005	Switch0	Server0	ICMP
	0.006	Laptop0	Switch2	ICMP
	0.006	PC1	Switch0	ICMP
	0.007	Switch2	Router 1	ICMP
	0.007	Switch0	Router4	ICMP
	0.008	Router 1	Router4	ICMP
	0.008	Router4	Router 1	ICMP
	0.009	Router4	Switch0	ICMP
	0.009	Router 1	Switch2	ICMP
	0.010	Switch0	PC0	ICMP
	0.010	Switch2	Laptop1	ICMP

ICMP events

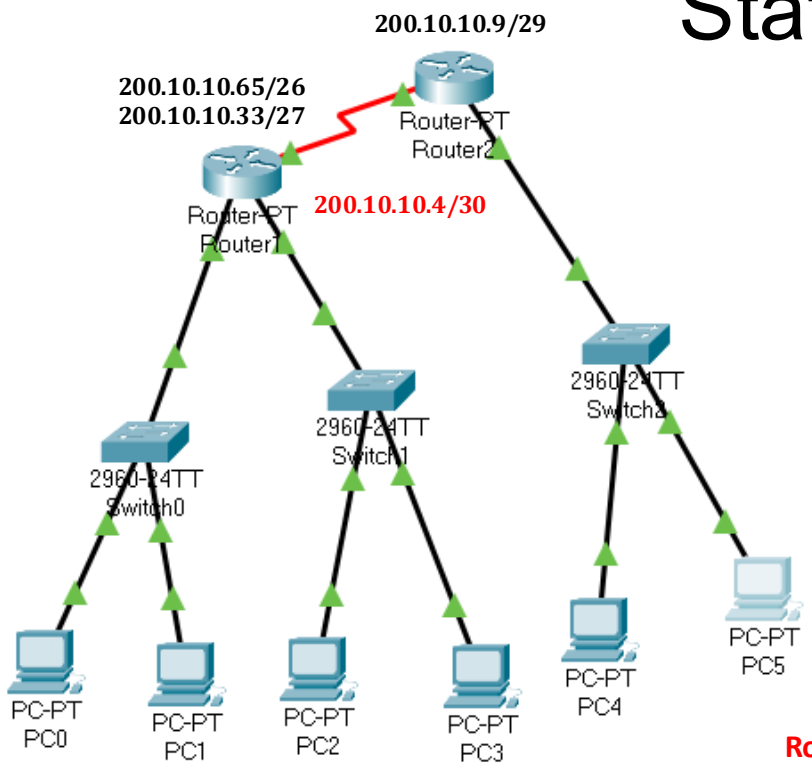
Fire	Last Status	Source	Destination	Type	Color	T
	Successful	PC0	Laptop0	ICMP		
	Successful	Laptop1	PC1	ICMP		

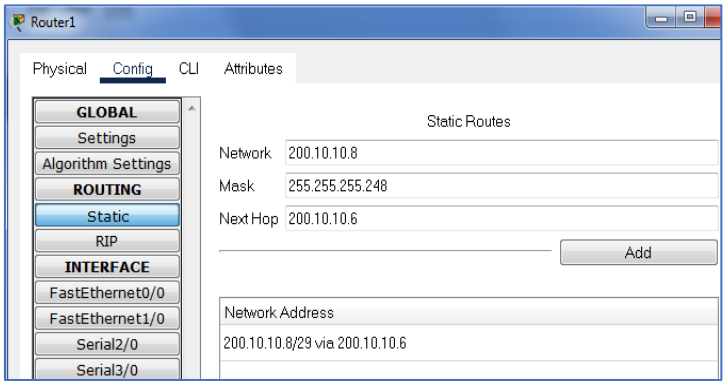
Simulation status

Host IP address for Laptops are manually assigned and static

Host IP address for PCs are dynamically assigned by DHCP

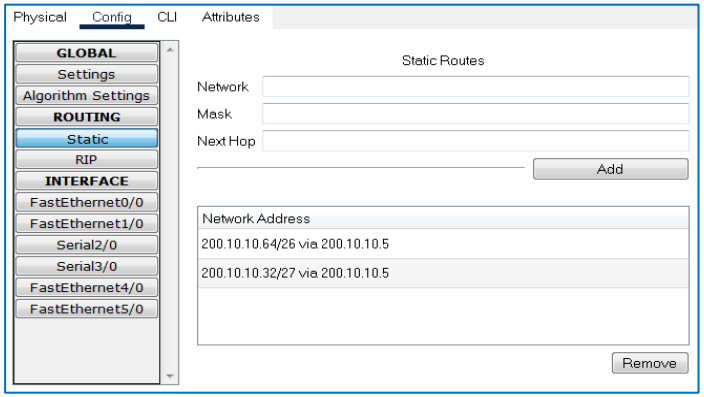
Static Routing





Configuring static route on Router 1

Static routes on Router 2



Routing Table 1

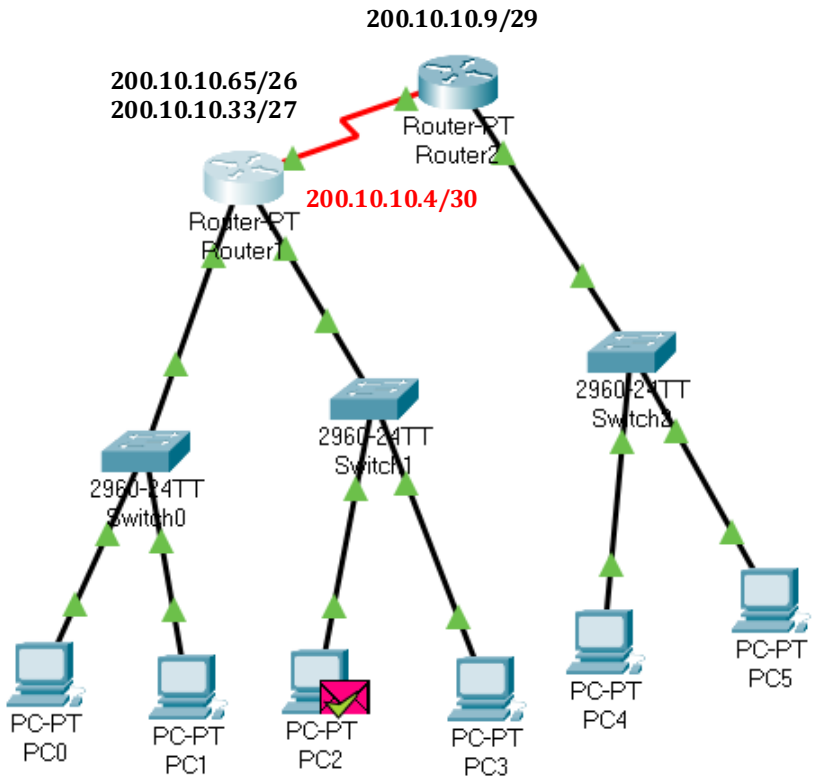
```
     200.10.10.0/24 is variably subnetted, 4 subnets, 4 masks
C       200.10.10.4/30 is directly connected, Serial2/0
S       200.10.10.8/29 [1/0] via 200.10.10.6
C       200.10.10.32/27 is directly connected, FastEthernet1/0
C       200.10.10.64/26 is directly connected, FastEthernet0/0
```

Router# show ip route

Routing Table 2
s = static
c = connected

```
     200.10.10.0/24 is variably subnetted, 4 subnets, 4 masks
C       200.10.10.4/30 is directly connected, Serial2/0
C       200.10.10.8/29 is directly connected, FastEthernet0/0
S       200.10.10.32/27 [1/0] via 200.10.10.5
S       200.10.10.64/26 [1/0] via 200.10.10.5
```

Simulation - Static Routing



ICMP events

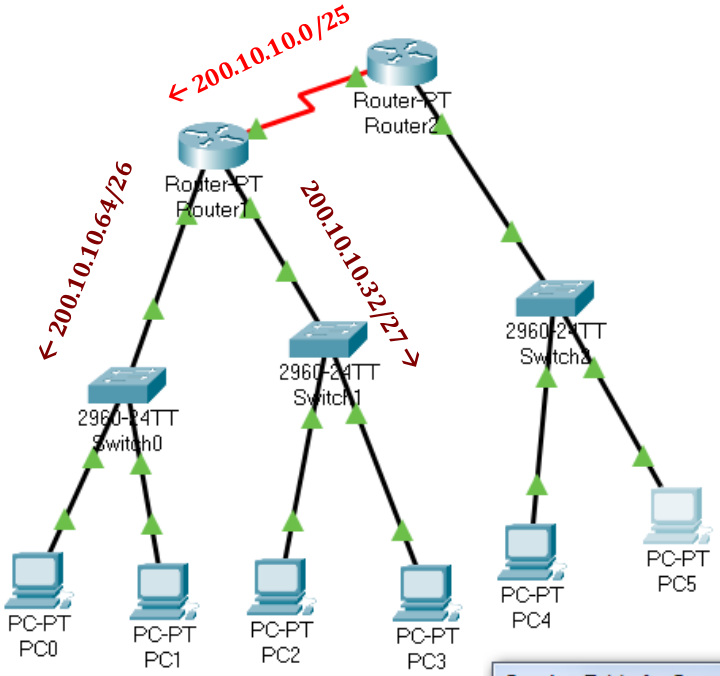
Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC0	ICMP
	0.000	–	PC2	ICMP
	0.001	PC0	Switch0	ICMP
	0.001	PC2	Switch1	ICMP
	0.002	Switch0	Router1	ICMP
	0.002	Switch1	Router1	ICMP
	0.003	Router1	Switch1	ICMP
	0.003	Router1	Router2	ICMP
	0.004	Switch1	PC2	ICMP
	0.004	Switch1	PC3	ICMP
	0.004	Router2	Switch2	ICMP
	0.005	PC3	Switch1	ICMP
	0.005	Switch2	PC4	ICMP
	0.005	Switch2	PC5	ICMP
	0.006	Switch1	Router1	ICMP
	0.006	PC4	Switch2	ICMP
	0.007	Router1	Switch0	ICMP
	0.007	Switch2	Router2	ICMP
	0.008	Switch0	PC0	ICMP
	0.008	Router2	Router1	ICMP
	0.009	Router1	Switch1	ICMP
	0.010	Switch1	PC2	ICMP

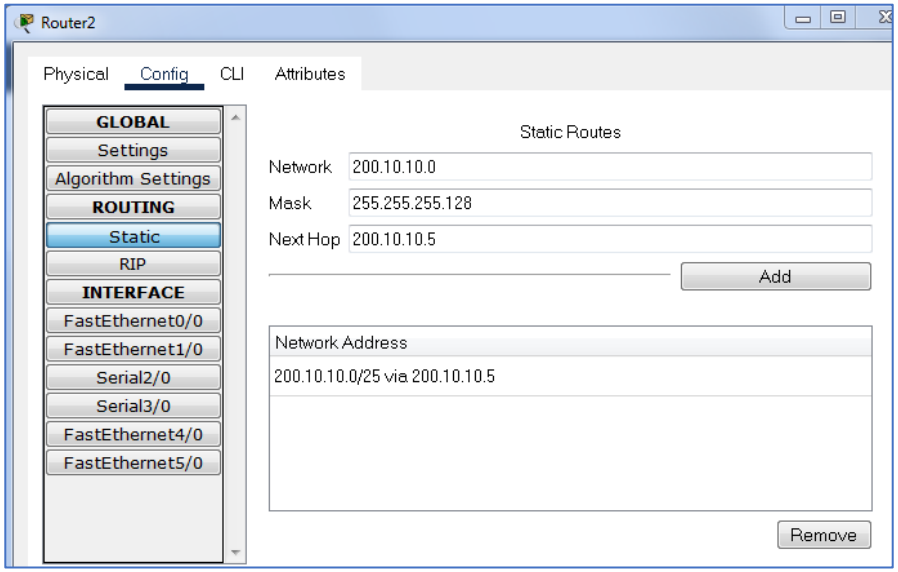
Simulation status

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC0	PC3	ICMP	
	Successful	PC2	PC4	ICMP	

Route Summarization



Configuring a supernet on Router 2



Routing Table for Router2

Type	Network	Port	Next Hop IP	Metric
S	200.10.10.0/25	—	200.10.10.5	1/0
C	200.10.10.4/30	Serial2/0	—	0/0
C	200.10.10.8/29	FastEthernet0/0	—	0/0

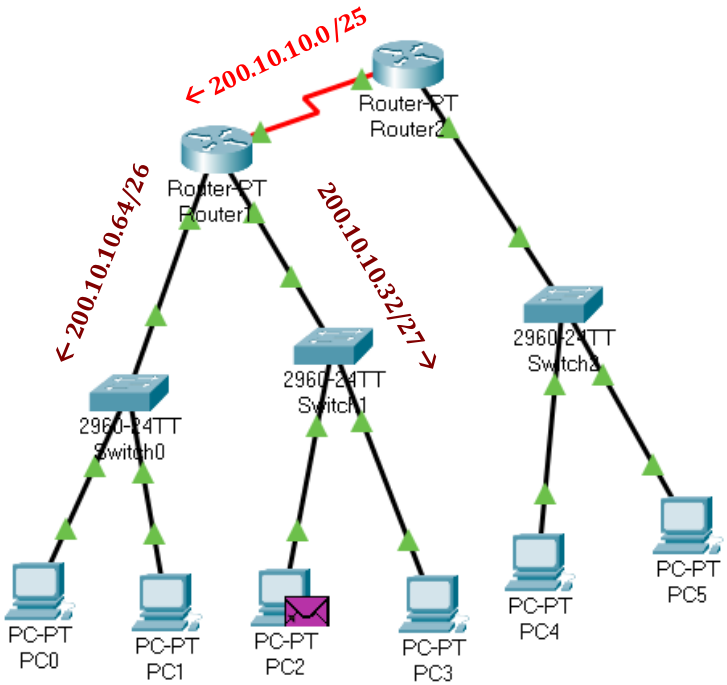
Summarization shorten the routing table

Routing Table for Router2

Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	—	0/0
C	200.10.10.8/29	FastEthernet0/0	—	0/0
S	200.10.10.32/27	—	200.10.10.5	1/0
S	200.10.10.64/26	—	200.10.10.5	1/0

Subnetting lengthen the routing table

Simulation - Route Summarization



ICMP event

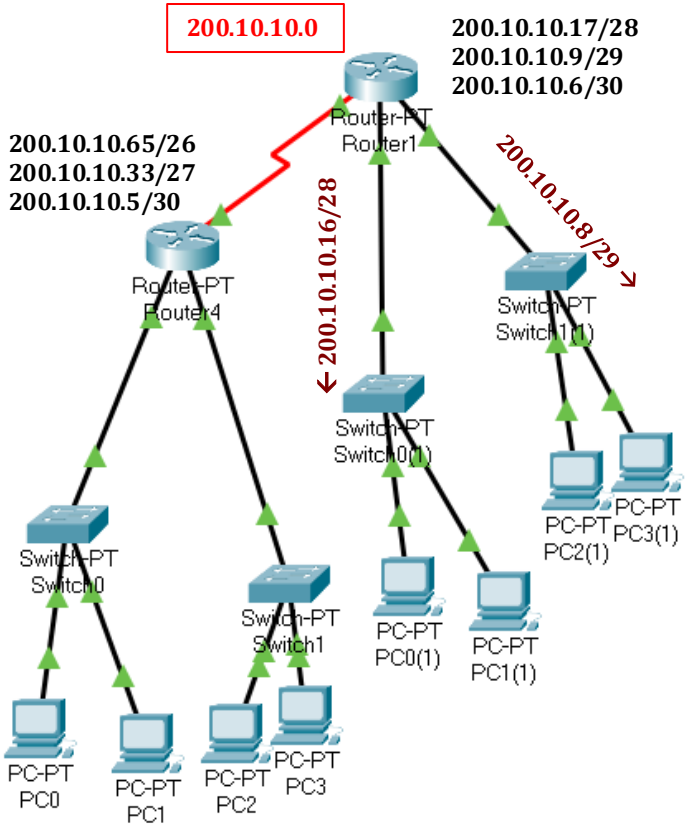
Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC5	ICMP
	0.001	PC5	Switch2	ICMP
	0.002	Switch2	Router2	ICMP
	0.003	Router2	Router1	ICMP
	0.004	Router1	Switch0	ICMP
	0.005	Switch0	PC0	ICMP
	0.006	PC0	Switch0	ICMP
	0.007	Switch0	Router1	ICMP
	0.008	Router1	Router2	ICMP
	0.009	Router2	Switch2	ICMP
	0.010	Switch2	PC5	ICMP
	0.010	–	PC2	ICMP
	0.011	PC2	Switch1	ICMP
	0.012	Switch1	Router1	ICMP
	0.013	Router1	Router2	ICMP
	0.014	Router2	Switch2	ICMP
	0.015	Switch2	PC4	ICMP
	0.015	Switch2	PC5	ICMP
	0.016	PC4	Switch2	ICMP
	0.017	Switch2	Router2	ICMP
	0.018	Router2	Router1	ICMP
	0.019	Router1	Switch1	ICMP
	0.020	Switch1	PC2	ICMP

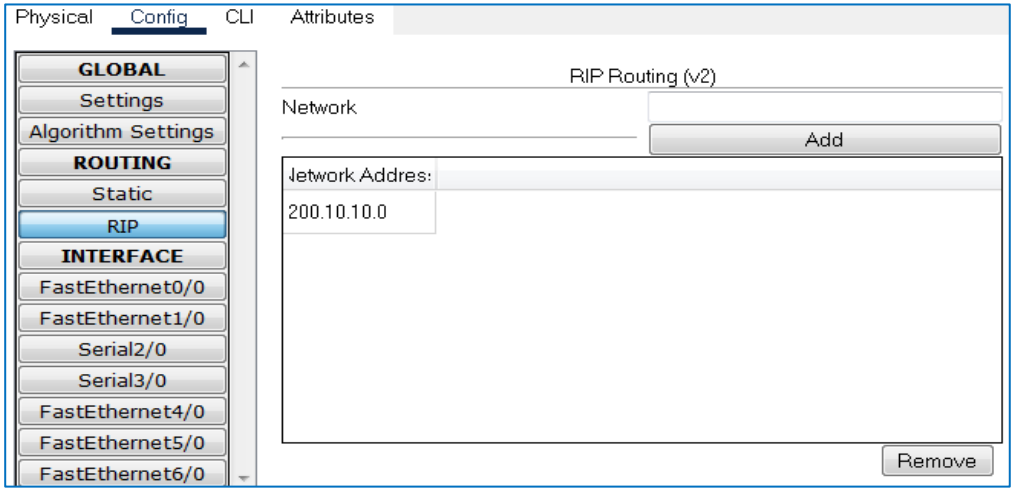
Simulation status

Fire	Last Status	Source	Destination	Type	Colc
	Successful	PC5	PC0	ICMP	
	Successful	PC2	PC4	ICMP	

Configuring RIP v2 (Dynamic Routing)



Add supernet 200.10.10.0 to rip table on Router 4



Show ip route:

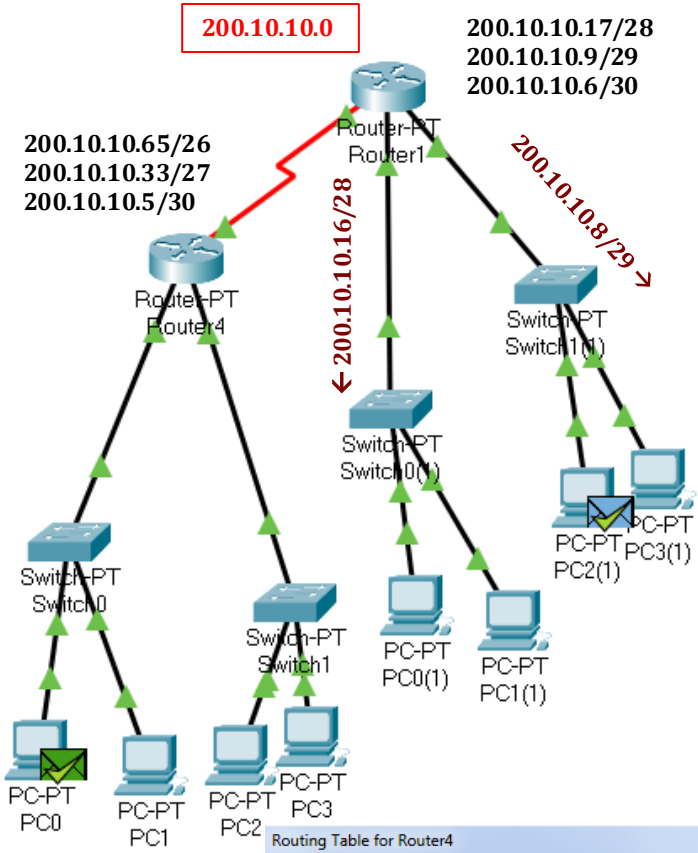
```
     200.10.10.0/24 is variably subnetted, 5 subnets, 5 masks
C       200.10.10.4/30 is directly connected, Serial2/0
R       200.10.10.8/29 [120/1] via 200.10.10.6, 00:00:04, Serial2/0
R       200.10.10.16/28 [120/1] via 200.10.10.6, 00:00:04, Serial2/0
C       200.10.10.32/27 is directly connected, FastEthernet1/0
C       200.10.10.64/26 is directly connected, FastEthernet0/0
```

Show running-config:

```
router rip
 version 2
 network 200.10.10.0
!
ip classless
!
```

- **Configure rip version 2**

Simulation - RIP v2 (Dynamic Routing)



Routing Table for Router1

Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	—	0/0
C	200.10.10.8/29	FastEthernet1/0	—	0/0
C	200.10.10.16/28	FastEthernet0/0	—	0/0
R	200.10.10.32/27	Serial2/0	200.10.10.5	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.5	120/1

Routing Table for Router4

Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	—	0/0
R	200.10.10.8/29	Serial2/0	200.10.10.6	120/1
R	200.10.10.16/28	Serial2/0	200.10.10.6	120/1
C	200.10.10.32/27	FastEthernet1/0	—	0/0
C	200.10.10.64/26	FastEthernet0/0	—	0/0

ICMP events

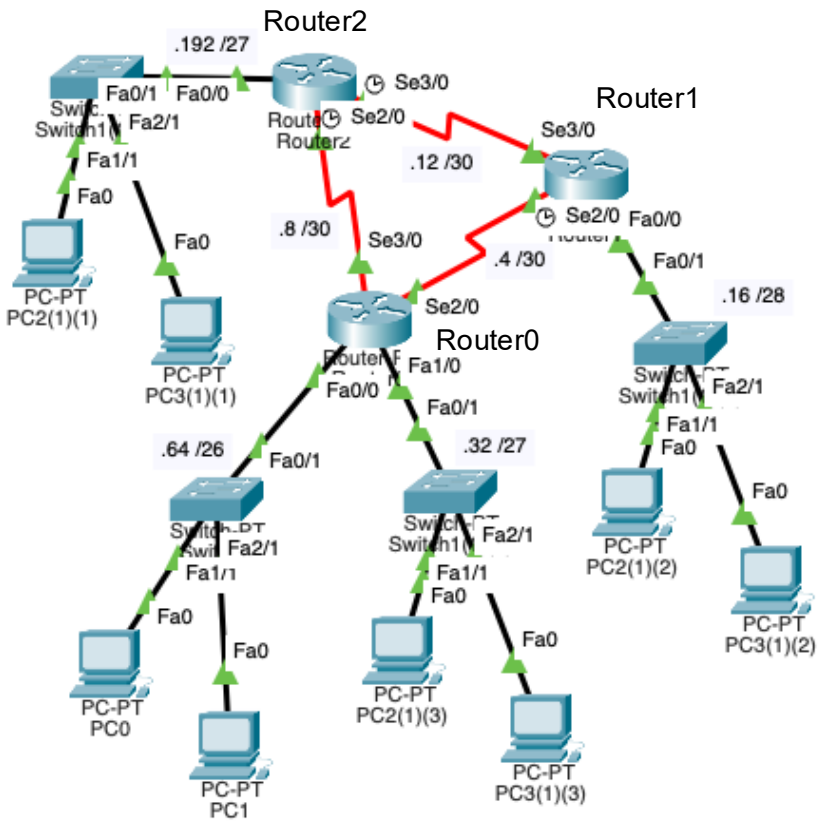
Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	–	PC0	ICMP
	0.000	–	PC2(1)	ICMP
	0.001	PC0	Switch0	ICMP
	0.001	PC2(1)	Switch1(1)	ICMP
	0.002	Switch0	Router4	ICMP
	0.002	Switch1(1)	Router1	ICMP
	0.003	Router4	Router1	ICMP
	0.003	Router1	Router4	ICMP
	0.004	Router1	Switch0(1)	ICMP
	0.004	Router4	Switch1	ICMP
	0.005	Switch0(1)	PC0(1)	ICMP
	0.005	Switch0(1)	PC1(1)	ICMP
	0.005	Switch1	PC2	ICMP
	0.005	Switch1	PC3	ICMP
	0.006	PC0(1)	Switch0(1)	ICMP
	0.006	PC2	Switch1	ICMP
	0.007	Switch0(1)	Router1	ICMP
	0.007	Switch1	Router4	ICMP
	0.008	Router1	Router4	ICMP
	0.008	Router4	Router1	ICMP
	0.009	Router4	Switch0	ICMP
	0.009	Router1	Switch1(1)	ICMP
	0.010	Switch0	PC0	ICMP
	0.010	Switch1(1)	PC2(1)	ICMP

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC0	PC0(1)	ICMP	
	Successful	PC2(1)	PC2	ICMP	

(No) Auto-Summary

Single network 200.10.10.0



Routing Table for Router2

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.4/30	Serial3/0	200.10.10.14	120/1
R	200.10.10.4/30	Serial2/0	200.10.10.10	120/1
C	200.10.10.8/30	Serial2/0	---	0/0
C	200.10.10.12/30	Serial3/0	---	0/0
R	200.10.10.16/28	Serial3/0	200.10.10.14	120/1
R	200.10.10.32/27	Serial2/0	200.10.10.10	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.10	120/1
C	200.10.10.192/27	FastEthernet0/0	---	0/0

Routing Table for Router0

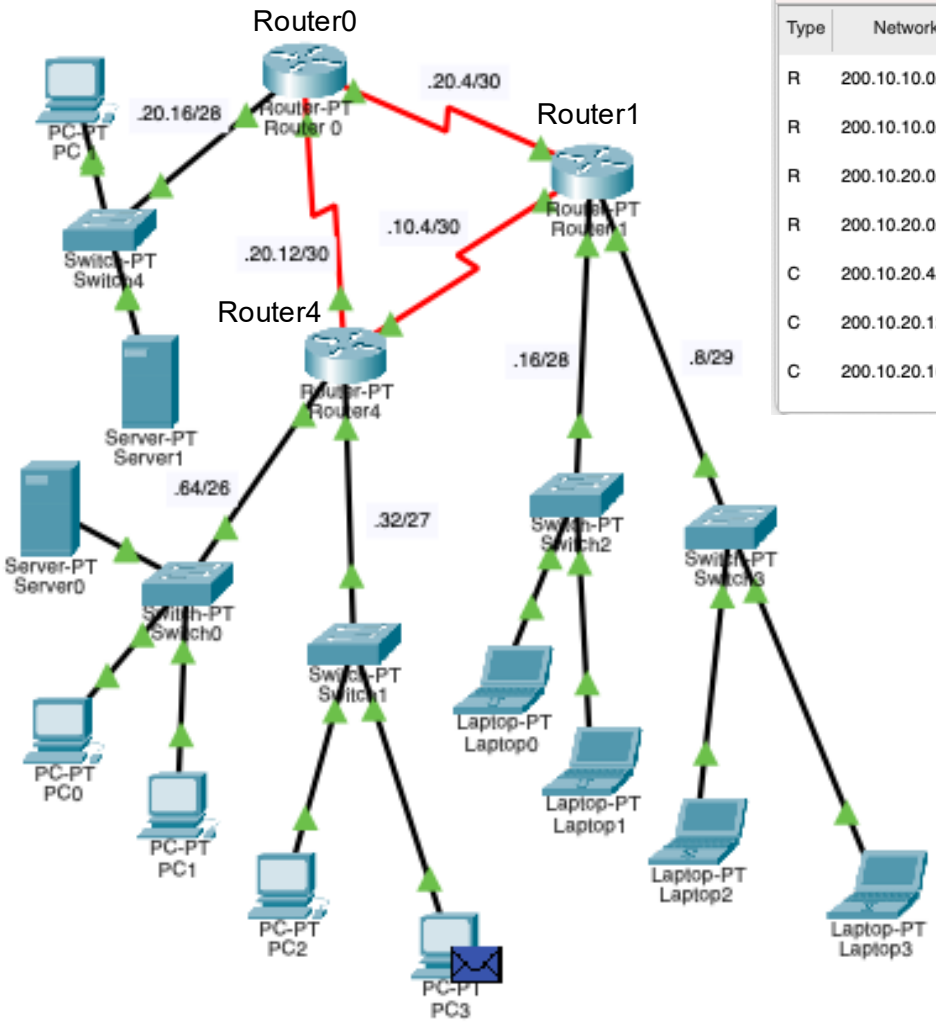
Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	---	0/0
C	200.10.10.8/30	Serial3/0	---	0/0
R	200.10.10.12/30	Serial3/0	200.10.10.9	120/1
R	200.10.10.12/30	Serial2/0	200.10.10.5	120/1
R	200.10.10.16/28	Serial2/0	200.10.10.5	120/1
C	200.10.10.32/27	FastEthernet1/0	---	0/0
C	200.10.10.64/26	FastEthernet0/0	---	0/0
R	200.10.10.192/27	Serial3/0	200.10.10.9	120/1

Routing Table for Router1

Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	---	0/0
R	200.10.10.8/30	Serial3/0	200.10.10.13	120/1
R	200.10.10.8/30	Serial2/0	200.10.10.6	120/1
C	200.10.10.12/30	Serial3/0	---	0/0
C	200.10.10.16/28	FastEthernet0/0	---	0/0
R	200.10.10.32/27	Serial2/0	200.10.10.6	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.6	120/1
R	200.10.10.192/27	Serial3/0	200.10.10.13	120/1

Auto-Summary

Two networks 200.10.10.0, 200.10.20.0



Routing Table for Router 0

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.10.20.6	120/1
R	200.10.10.0/24	Serial2/0	200.10.20.14	120/1
R	200.10.20.0/24	Serial2/0	200.10.20.14	120/2
R	200.10.20.0/24	Serial3/0	200.10.20.6	120/2
C	200.10.20.4/30	Serial3/0	---	0/0
C	200.10.20.12/30	Serial2/0	---	0/0
C	200.10.20.16/28	FastEthernet0/0	---	0/0

Routing Table for Router 1

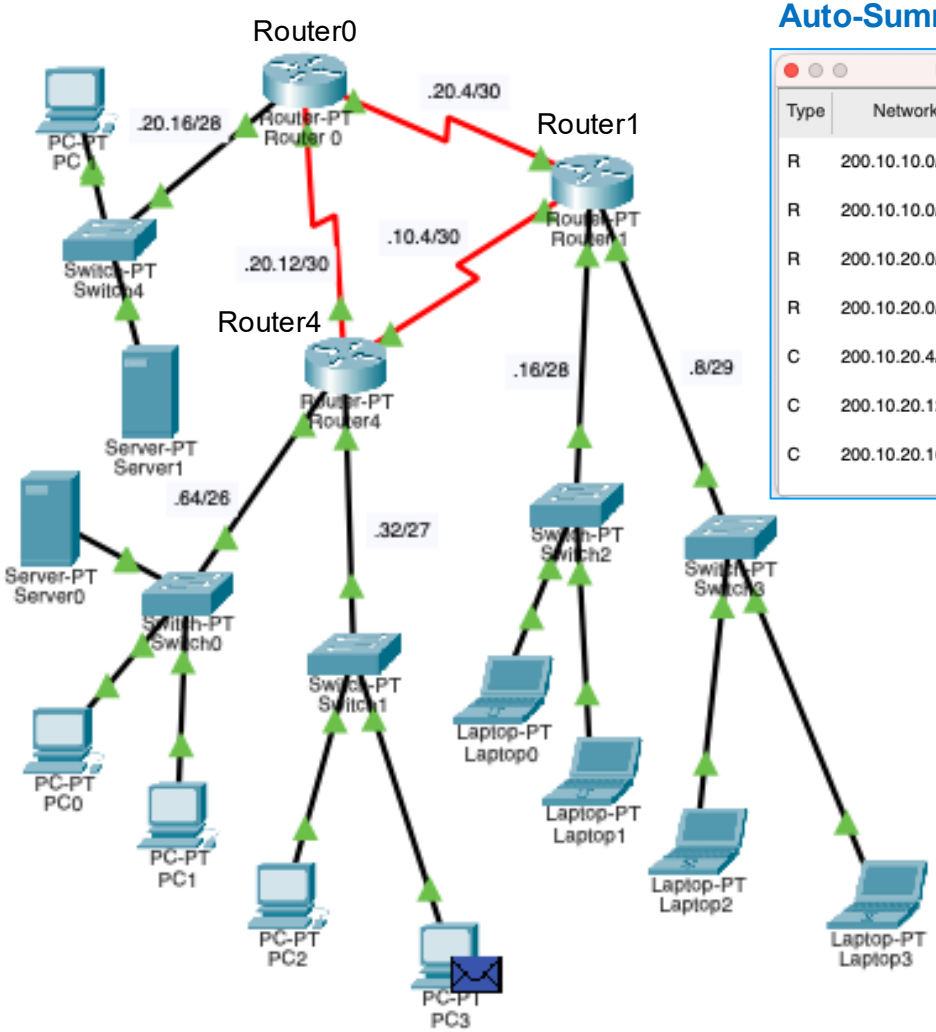
Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	---	0/0
C	200.10.10.8/29	FastEthernet1/0	---	0/0
C	200.10.10.16/28	FastEthernet0/0	---	0/0
R	200.10.10.32/27	Serial2/0	200.10.10.5	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.5	120/1
R	200.10.20.0/24	Serial2/0	200.10.10.5	120/1
C	200.10.20.4/30	Serial3/0	---	0/0
R	200.10.20.12/30	Serial3/0	200.10.20.5	120/1
R	200.10.20.16/28	Serial3/0	200.10.20.5	120/1

Routing Table for Router4

Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	---	0/0
R	200.10.10.8/29	Serial2/0	200.10.10.6	120/1
R	200.10.10.16/28	Serial2/0	200.10.10.6	120/1
C	200.10.10.32/27	FastEthernet1/0	---	0/0
C	200.10.10.64/26	FastEthernet0/0	---	0/0
R	200.10.20.0/24	Serial2/0	200.10.10.6	120/1
R	200.10.20.4/30	Serial3/0	200.10.20.13	120/1
C	200.10.20.12/30	Serial3/0	---	0/0
R	200.10.20.16/28	Serial3/0	200.10.20.13	120/1

No Auto-Summary

Two networks 200.10.10.0, 200.10.20.0



Auto-Summary on Router 0

Routing Table for Router 0

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.10.20.6	120/1
R	200.10.10.0/24	Serial2/0	200.10.20.14	120/1
R	200.10.20.0/24	Serial2/0	200.10.20.14	120/2
R	200.10.20.0/24	Serial3/0	200.10.20.6	120/2
C	200.10.20.4/30	Serial3/0	---	0/0
C	200.10.20.12/30	Serial2/0	---	0/0
C	200.10.20.16/28	FastEthernet0/0	---	0/0

No Auto-Summary on Router 0

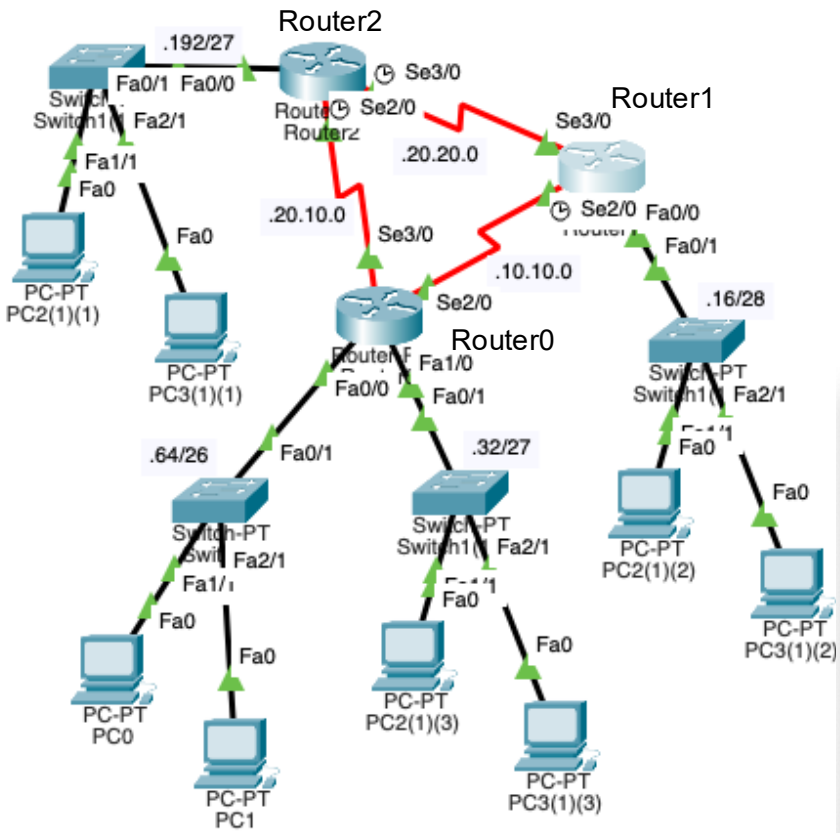
Routing Table for Router 0

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial2/0	200.10.20.14	120/1
R	200.10.10.0/24	Serial3/0	200.10.20.6	120/1
R	200.10.10.8/29	Serial2/0	200.10.20.14	120/2
R	200.10.10.16/28	Serial2/0	200.10.20.14	120/2
R	200.10.10.32/27	Serial3/0	200.10.20.6	120/2
R	200.10.10.64/26	Serial3/0	200.10.20.6	120/2
R	200.10.20.0/24	Serial3/0	200.10.20.6	120/2
R	200.10.20.0/24	Serial2/0	200.10.20.14	120/2
C	200.10.20.4/30	Serial3/0	---	0/0
C	200.10.20.12/30	Serial2/0	---	0/0
C	200.10.20.16/28	FastEthernet0/0	---	0/0

```
router rip
 version 2
 network 200.10.20.0
 no auto-summary
!
```

Auto-Summary

Multiple networks 200… .10.10.0, .20.20.0, .20.10.0, .10.20.0



Routing Table for Router2

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.20.20.14	120/1
R	200.10.10.0/24	Serial2/0	200.20.10.10	120/1
C	200.10.10.192/27	FastEthernet0/0	---	0/0
C	200.20.10.8/30	Serial2/0	---	0/0
C	200.20.20.12/30	Serial3/0	---	0/0

Routing Table for Router0

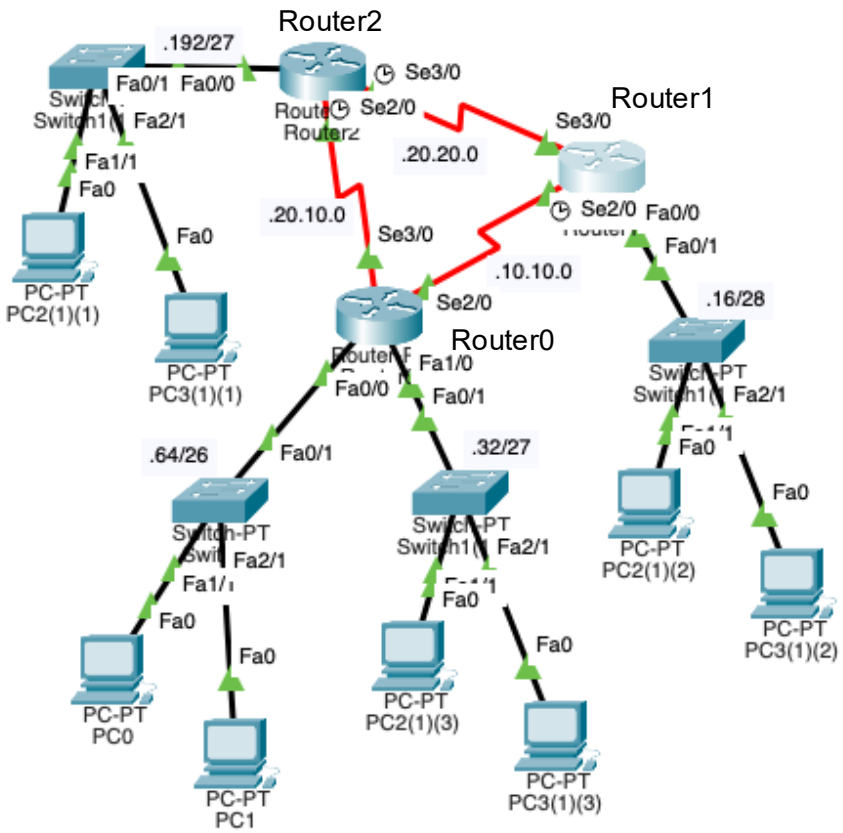
Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.20.10.9	120/1
C	200.10.10.4/30	Serial2/0	---	0/0
R	200.10.10.16/28	Serial2/0	200.10.10.5	120/1
C	200.10.10.32/27	FastEthernet1/0	---	0/0
C	200.10.10.64/26	FastEthernet0/0	---	0/0
C	200.20.10.8/30	Serial3/0	---	0/0
R	200.20.20.0/24	Serial3/0	200.20.10.9	120/1
R	200.20.20.0/24	Serial2/0	200.10.10.5	120/1

Routing Table for Router1

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.20.20.13	120/1
C	200.10.10.4/30	Serial2/0	---	0/0
C	200.10.10.16/28	FastEthernet0/0	---	0/0
R	200.10.10.32/27	Serial2/0	200.10.10.6	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.6	120/1
R	200.20.10.0/24	Serial2/0	200.10.10.6	120/1
R	200.20.10.0/24	Serial3/0	200.20.20.13	120/1
C	200.20.20.12/30	Serial3/0	---	0/0

No Auto-Summary

Multiple networks 200… .10.10.0, .20.20.0, .20.10.0, .10.20.0



Router2

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial2/0	200.20.10.10	120…
R	200.10.10.0/24	Serial3/0	200.20.20.14	120…
R	200.10.10.4/30	Serial3/0	200.20.20.14	120/1
R	200.10.10.4/30	Serial2/0	200.20.10.10	120/1
R	200.10.10.16/28	Serial3/0	200.20.20.14	120/1
R	200.10.10.32/27	Serial2/0	200.20.10.10	120/1
R	200.10.10.64/26	Serial2/0	200.20.10.10	120/1
C	200.10.10.192/27	FastEthernet0/0	---	0/0
R	200.20.10.0/24	Serial2/0	200.20.10.10	120…
C	200.20.10.8/30	Serial2/0	---	0/0
C	200.20.20.12/30	Serial3/0	---	0/0

Routing Table for Router0

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.20.10.9	120…
C	200.10.10.4/30	Serial2/0	---	0/0
R	200.10.10.16/28	Serial2/0	200.10.10.5	120/1
C	200.10.10.32/27	FastEthernet1/0	---	0/0
C	200.10.10.64/26	FastEthernet0/0	---	0/0
R	200.10.10.19…	Serial3/0	200.20.10.9	120/1
R	200.20.10.0/24	Serial2/0	200.10.10.5	120…
C	200.20.10.8/30	Serial3/0	---	0/0
R	200.20.20.0/24	Serial3/0	200.20.10.9	120…
R	200.20.20.0/24	Serial2/0	200.10.10.5	120…
R	200.20.20.12/30	Serial2/0	200.10.10.5	120/1
R	200.20.20.12/30	Serial3/0	200.20.10.9	120/1

Routing Table for Router1

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.20.20.13	120…
C	200.10.10.4/30	Serial2/0	---	0/0
C	200.10.10.16/28	FastEthernet0/0	---	0/0
R	200.10.10.32/27	Serial2/0	200.10.10.6	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.6	120/1
R	200.10.10.19…	Serial3/0	200.20.20.13	120/1
R	200.20.10.0/24	Serial3/0	200.20.20.13	120…
R	200.20.10.8/30	Serial2/0	200.10.10.6	120/1
R	200.20.10.8/30	Serial3/0	200.20.20.13	120/1
C	200.20.20.12/30	Serial3/0	---	0/0

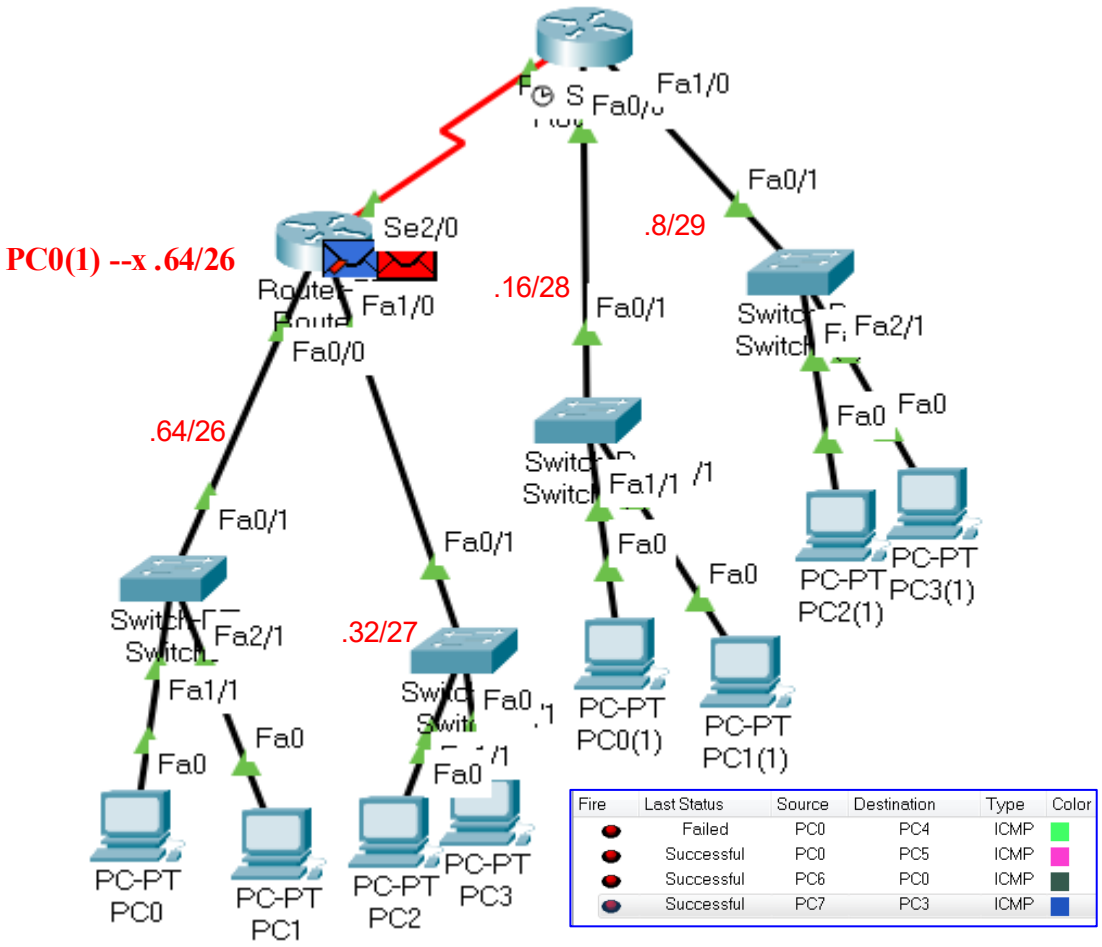
Last Status | Source | Destination | Type

Standard ACL

```
interface FastEthernet0/0
 ip address 200.10.10.17 255.255.255.240
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.9 255.255.255.248
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.6 255.255.255.252
 clock rate 2000000
!
```

```
interface FastEthernet0/0
 ip address 200.10.10.65 255.255.255.192
 ip access-group 1 out
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.33 255.255.255.224
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.5 255.255.255.252
!
```

```
!
access-list 1 deny host 200.10.10.18
access-list 1 permit any
!
```



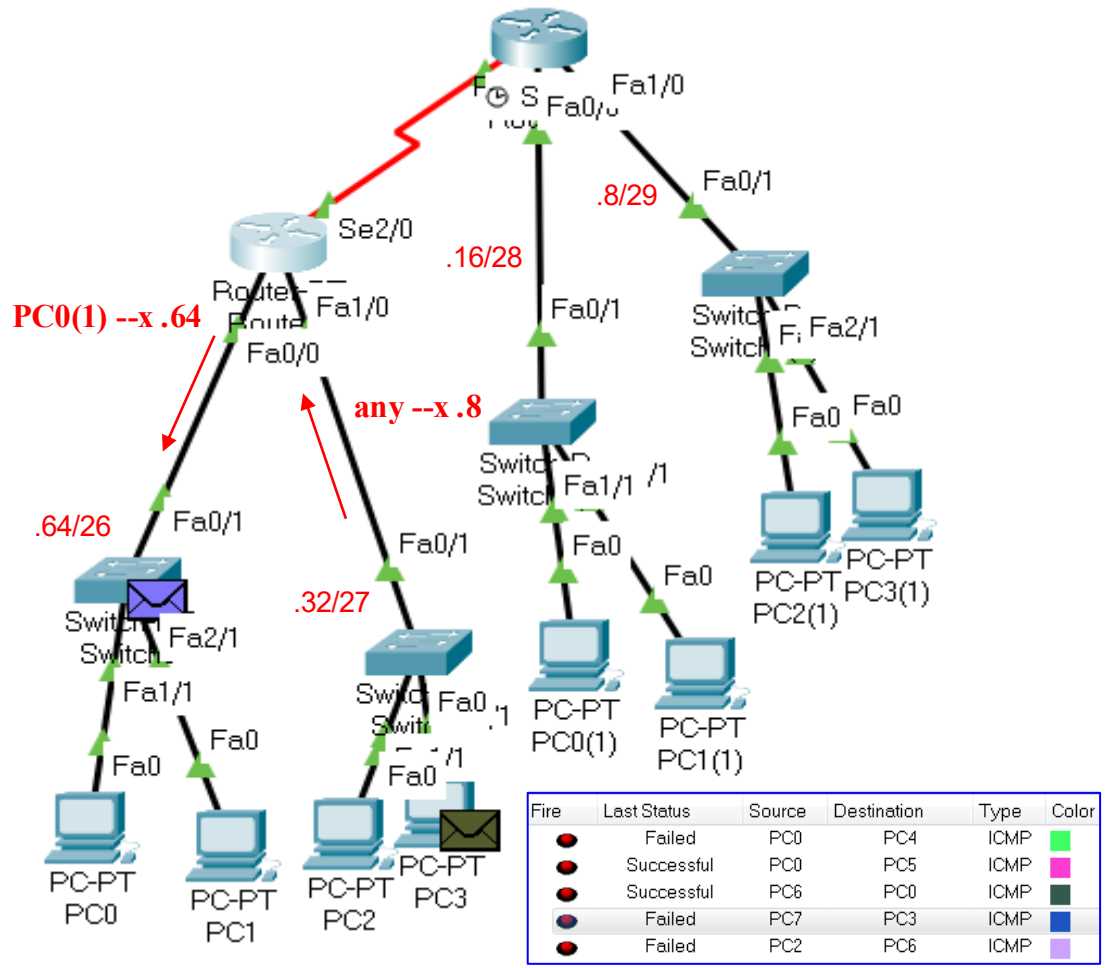
Fire	Last Status	Source	Destination	Type	Color
	Failed	PC0	PC4	ICMP	
	Successful	PC0	PC5	ICMP	
	Successful	PC6	PC0	ICMP	
	Successful	PC7	PC3	ICMP	

Extended ACL - IP

```
interface FastEthernet0/0
 ip address 200.10.10.17 255.255.255.240
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.9 255.255.255.248
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.6 255.255.255.252
 clock rate 2000000
!
```

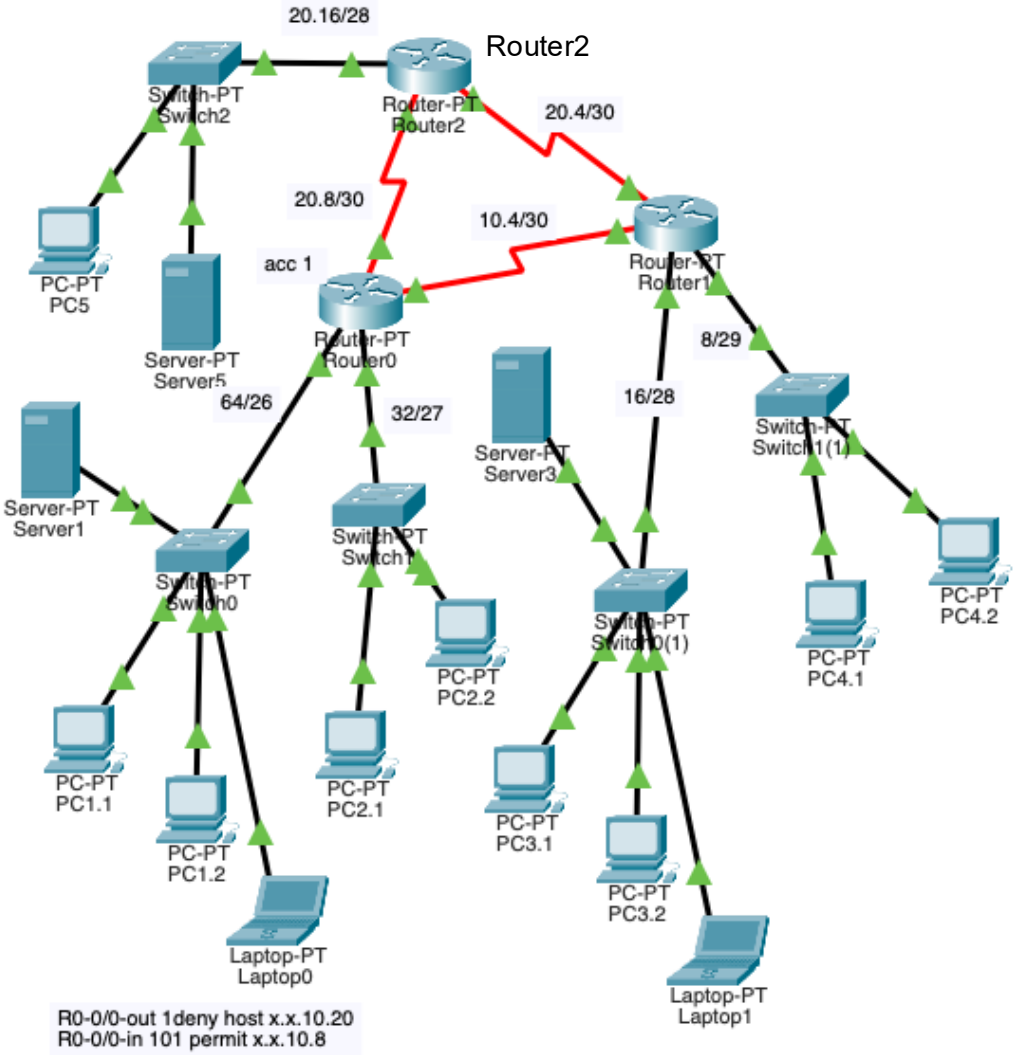
```
!
interface FastEthernet0/0
 ip address 200.10.10.65 255.255.255.192
 ip access-group 1 out
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.33 255.255.255.224
 ip access-group 101 in
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.5 255.255.255.252
!
```

```
access-list 1 deny host 200.10.10.18
access-list 1 permit any
access-list 101 deny ip any 200.10.10.8 0.0.0.7
access-list 101 permit ip any any
!
```



Fire	Last Status	Source	Destination	Type	Color
	Failed	PC0	PC4	ICMP	
	Successful	PC0	PC5	ICMP	
	Successful	PC6	PC0	ICMP	
	Failed	PC7	PC3	ICMP	
	Failed	PC2	PC6	ICMP	

Extended ACL – TCP




```
Router2,  interfaee fast 0/0
ip address 200.10.20.17 255.255.255.240
ip access-group 105 out
duplex auto
speed auto
!
```

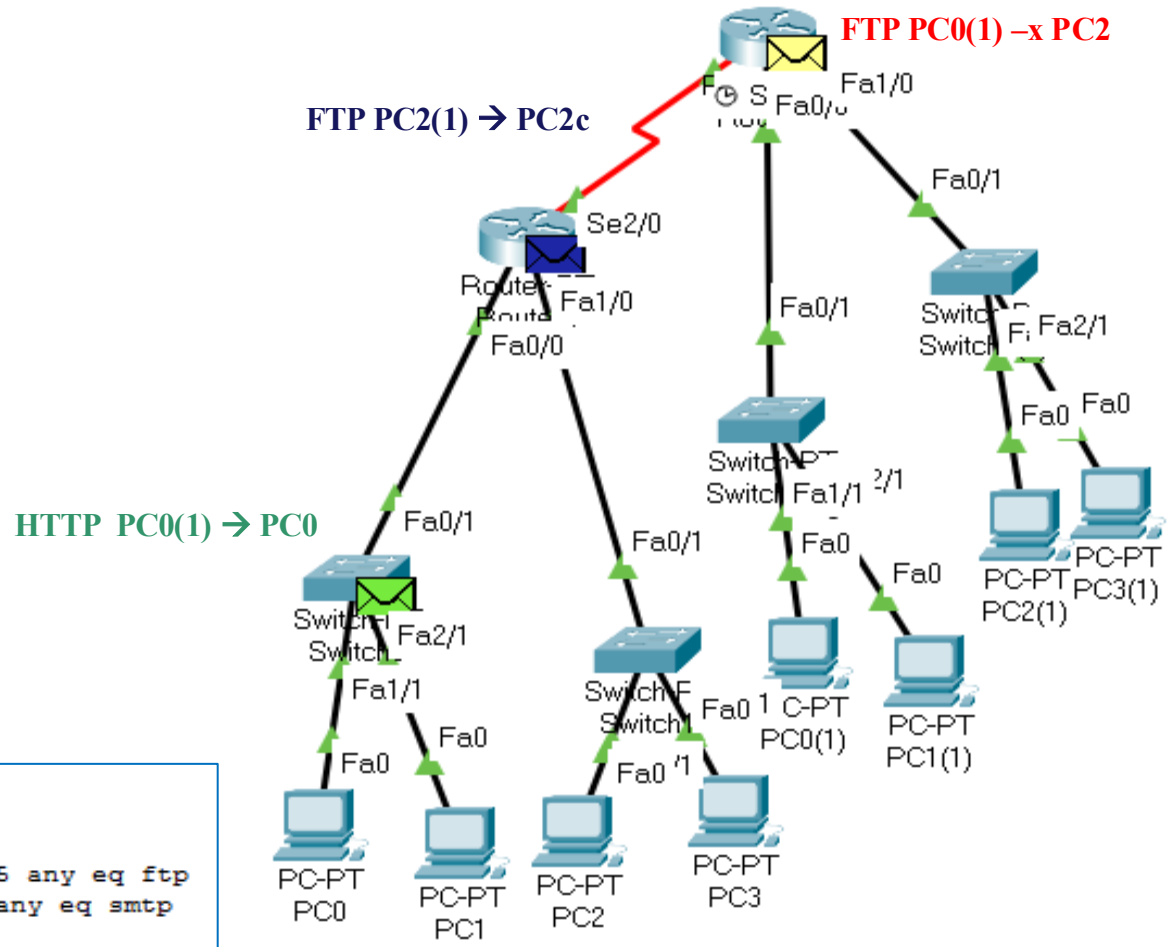
```
access-list 102 permit tcp 200.10.10.0 0.0.0.31 any eq ftp
access-list 102 permit tcp any any

access-list 101 permit tcp 200.10.10.0 0.0.0.31 any eq ftp
access-list 101 permit ip any any

access-list 103 permit tcp 200.10.10.0 0.0.0.31 any eq ftp
access-list 103 permit tcp 200.10.10.0 0.0.0.127 any eq www

access-list 104 permit tcp 200.10.10.64 0.0.0.63 any eq www
access-list 104 permit tcp 200.10.10.0 0.0.0.31 any eq ftp

access-list 105 deny tcp 200.10.10.0 0.0.0.31 any eq ftp
access-list 105 deny tcp 200.10.10.64 0.0.0.31 any eq www
access-list 105 permit ip any any
!
```

Extended ACL – TCP(1)

```
interface FastEthernet0/0
 ip address 200.10.10.17 255.255.255.240
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.9 255.255.255.248
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.6 255.255.255.252
 clock rate 2000000
!
```

```
!
interface FastEthernet0/0
 ip address 200.10.10.65 255.255.255.192
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.33 255.255.255.224
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.5 255.255.255.252
 ip access-group 101 in
!
```

```
!
access-list 1 deny host 200.10.10.18
access-list 1 permit any
access-list 101 deny tcp 200.10.10.16 0.0.0.15 any eq ftp
access-list 101 deny tcp 200.10.10.8 0.0.0.7 any eq smtp
access-list 101 permit ip any any
!
```



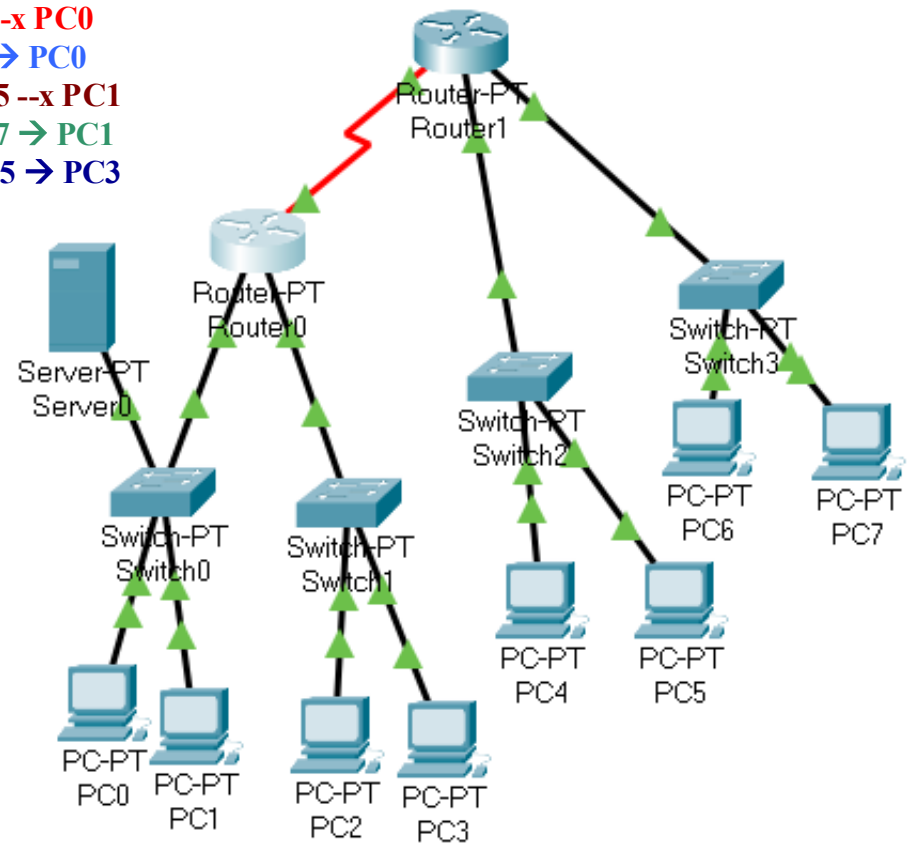
Extended ACL – TCP(2)

```
interface FastEthernet0/0
 ip address 200.10.10.17 255.255.255.240
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.9 255.255.255.248
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.6 255.255.255.252
 clock rate 2000000
!
```

```
interface FastEthernet0/0
 ip address 200.10.10.65 255.255.255.192
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 200.10.10.33 255.255.255.224
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 200.10.10.6 255.255.255.252
 ip access-group 102 in
```

```
access-list 102 deny tcp 200.10.10.16 0.0.0.15 any eq ftp
access-list 102 deny tcp 200.10.10.8 0.0.0.7 any eq smtp
access-list 102 permit ip any any
```

FTP PC4 --x PC0
FTP PC6 → PC0
SMTP PC5 --x PC1
SMTP PC7 → PC1
HTTP PC5 → PC3



Default directory for .pkt files saved

- Default directory:
  - C:\Users\<user_name>\Cisco Packet Tracer 8.2\saves